Generative Adversarial Networks for Synthetic Data Generation

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GitHub Repository: GANs Repository

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1. Introduction

Generative Artificial Intelligence is one of the fastest-growing fields in machine learning. Among the generative approaches, Generative Adversarial Networks (GANs) have been widely explored due to their ability to generate realistic synthetic data. The first introduction of GANs by Ian Goodfellow et al. (2014) describes two neural networks that compete against each other during training: a Generator and a Discriminator.

In a Generative Adversarial Network (GAN), two neural networks work against each other during training. The Generator is responsible for creating synthetic samples from random noise, while the Discriminator evaluates these samples and attempts to determine whether they are real or artificially generated. As training progresses, the Generator gradually improves its ability to produce more realistic outputs by learning from the Discriminator's feedback. This competitive learning process enables GANs to generate high-quality synthetic data that closely resembles real-world observations. Due to their effectiveness, GANs have been widely applied across various fields, including medical image analysis, cybersecurity, image generation, data augmentation, and creative content production.

The aim of this assignment was to study the effectiveness of GAN's in multiple domains. In the first part, we focused on understanding the fundamentals of GANs using synthetic 2D data. The second part focused on three practical applications: medical imaging, cyber security and creative sketch generation. All experiments were implemented with Pytorch in Google Colab

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2. Part 1: Building and Understanding GANs from Scratch

2.1 Sine-Wave GAN

The first experiment was a replication of the sine-wave GAN shown in the practical tutorial. A synthetic dataset of 2,000 samples was generated by using a sine function with added Gaussian noise. Each sample had two features, corresponding to the x and y coordinates.

The Generator was built using fully connected neural network layers with ReLU activation functions. The Discriminator employed LeakyReLU activations and a sigmoid output layer. The training was performed using the binary cross entropy loss and the Adam optimiser.

The GAN was able to learn the overall shape of the sine-wave distribution. By visual inspection we see that the points generated gradually match the real distribution as training progresses.

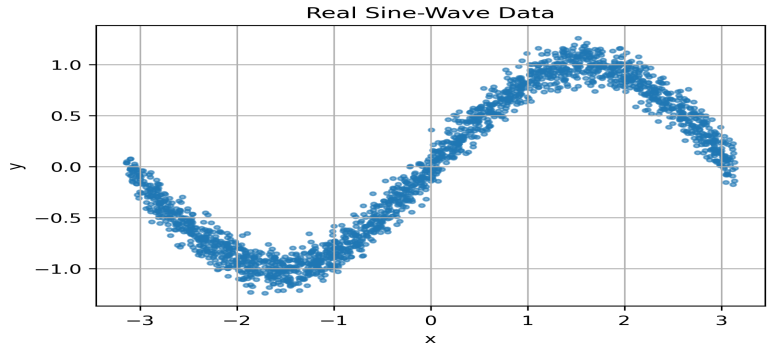

Figure 1: Real Sine-Wave Dataset

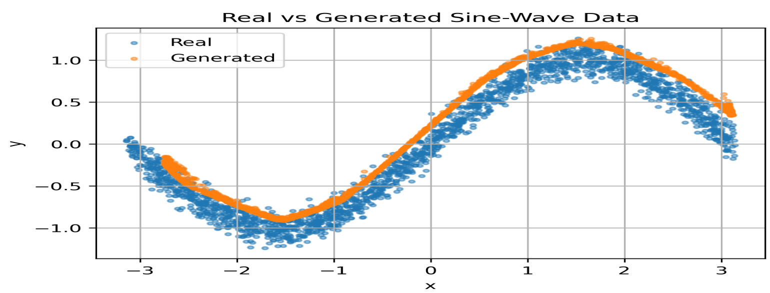

Figure 2: Real vs Generated Sine-Wave Samples

2.2 Noisy Parametric Curve

To increase the complexity of the learning task, a second synthetic distribution was generated using:

$y = \sin(2x) + 0.3\cos(5x) + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$ is Gaussian noise.

This is a more complicated distribution than a simple sine wave, with multiple oscillations and local variations. First, the original architecture of the GAN was trained on this dataset, and then a modified architecture was introduced.

The modified GAN added more hidden layers, Batch Normalization and LeakyReLU activation functions. With these changes, training became more stable and the samples generated were closer to the underlying distribution.

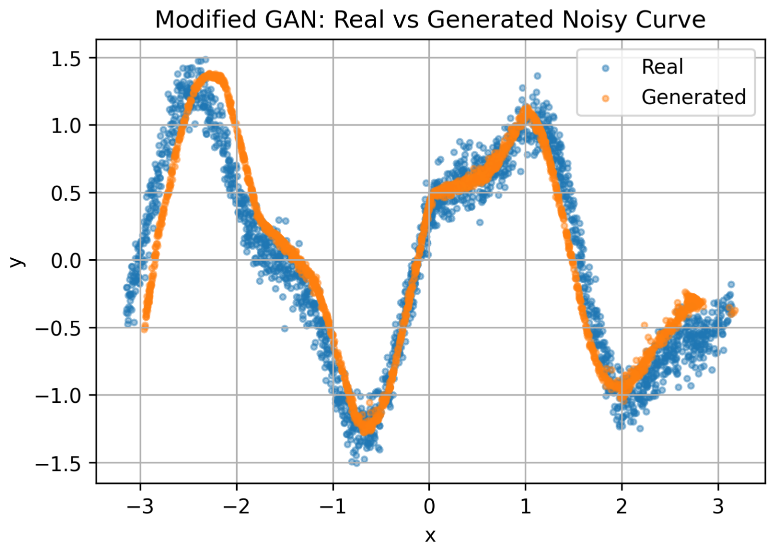

Figure 3: Original vs Modified GAN on Noisy Parametric Curve

The results demonstrated that architectural choices can significantly influence GAN performance. The modified model showed better convergence and generated more realistic samples than the baseline implementation.

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3. Part 2.1: Medical Imaging Using PneumoniaMNIST

Dataset Exploration

The medical imaging experiment used the PneumoniaMNIST dataset from MedMNIST. The dataset consists of 28×28 grayscale chest X-ray images categorised into normal and pneumonia classes.

The training set contained 4,708 images while the test set contained 624 images. Analysis of the class distribution showed that pneumonia cases were more common than normal cases, with 3,494 pneumonia images and 1,214 normal images.

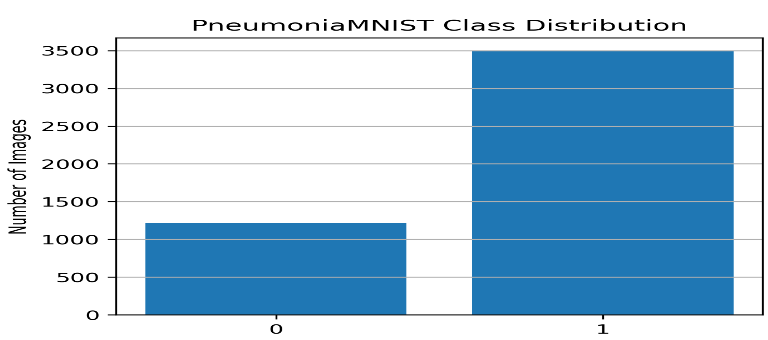

Figure 4: PneumoniaMNIST Class Distribution

DCGAN Architecture

This assignment implemented a Deep Convolutional Generative Adversarial Network (DCGAN), following the architecture proposed by Alec Radford and colleagues (2015). The Generator network was designed to convert random noise vectors into synthetic chest X-ray images using a series of transposed convolution layers. At the same time, the Discriminator network analysed both real and generated images and attempted to determine whether each image was authentic or artificially created.

The model was trained using the Adam optimisation algorithm with a learning rate of 0.0002, a setting commonly adopted in GAN research due to its stable training performance. During the training process, both Generator and Discriminator loss values were recorded and monitored to evaluate learning progress and assess whether the model was converging towards a stable solution. Tracking these losses also provided insight into the balance between the two networks and the overall effectiveness of the adversarial training process.

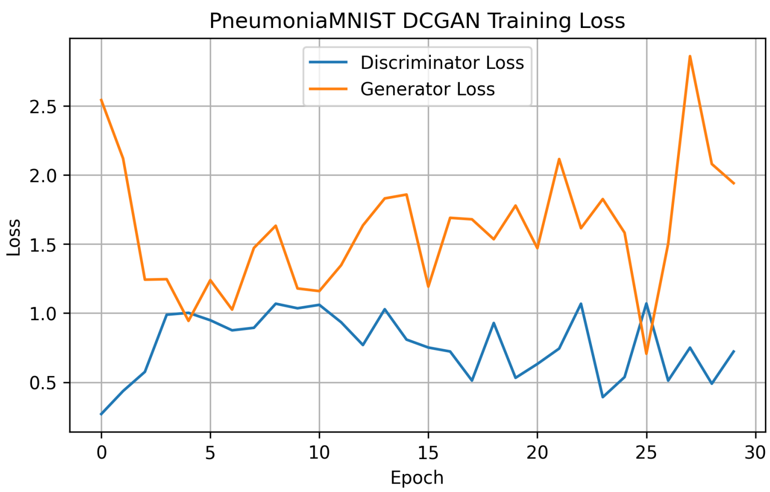

Figure 5: PneumoniaMNIST DCGAN Loss Curves

Results and Discussion

The generated images displayed structures that resembled chest X-rays, indicating that the Generator successfully learned important characteristics of the dataset. However, some generated images contained visible artefacts and lacked the fine anatomical details present in real medical images.

A simplified FID-style score of **62.37** was obtained. Lower scores indicate greater similarity between real and generated distributions. Pixel statistics also showed reasonable similarity between real and generated samples, with mean pixel values of 0.164 and 0.127 respectively.

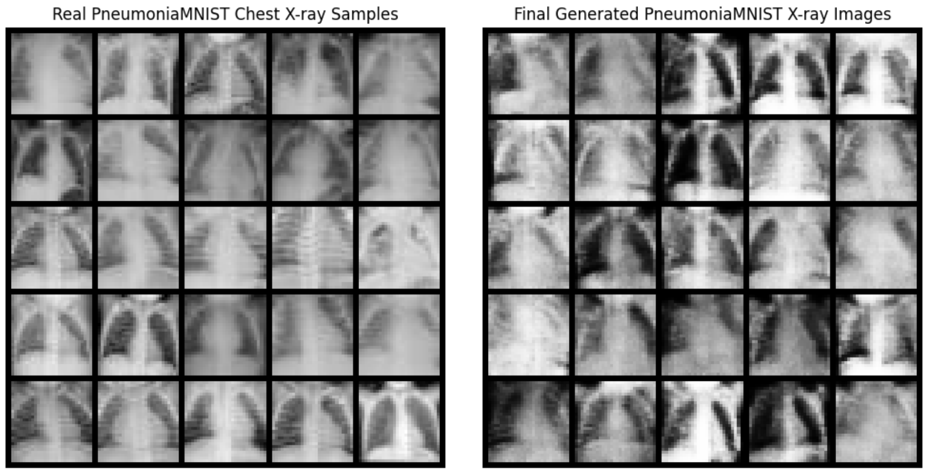

Figure 6: Real vs Generated Chest X-Ray Samples

The same DCGAN approach generalises beyond medical imaging, a GAN trained on duck photographs could similarly learn to generate new duck images that preserve shape, texture and colour. However, natural images of this kind are typically harder to model than chest X-rays because of variation in lighting, background and colour, whereas medical scans are grayscale and captured under controlled conditions.

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4. Part 2.2: Cybersecurity – Synthetic Network Traffic Generation

Dataset Exploration

The cybersecurity experiment utilized the Wednesday subset of the CICIDS2017 dataset developed by the Canadian Institute for Cybersecurity. The dataset contains network traffic labelled as benign or malicious.

Only BENIGN and DoS-related records were retained for training. Numerical features were standardized, while missing and infinite values were removed to ensure data quality.

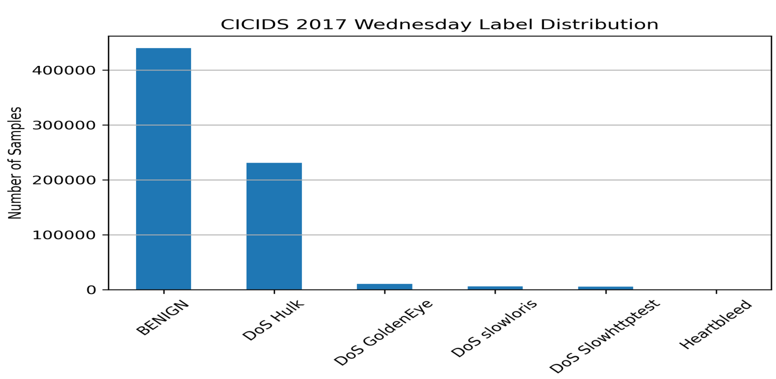

Figure 7: CICIDS2017 Label Distribution

Tabular GAN

Unlike image-based GANs, network traffic data consists of numerical feature vectors rather than pixels. Therefore, a fully connected GAN architecture was implemented to model the feature distributions.

After training, synthetic traffic samples were generated and compared with real traffic using Principal Component Analysis (PCA). PCA enables high-dimensional data to be visualized in two dimensions while preserving the most important patterns.

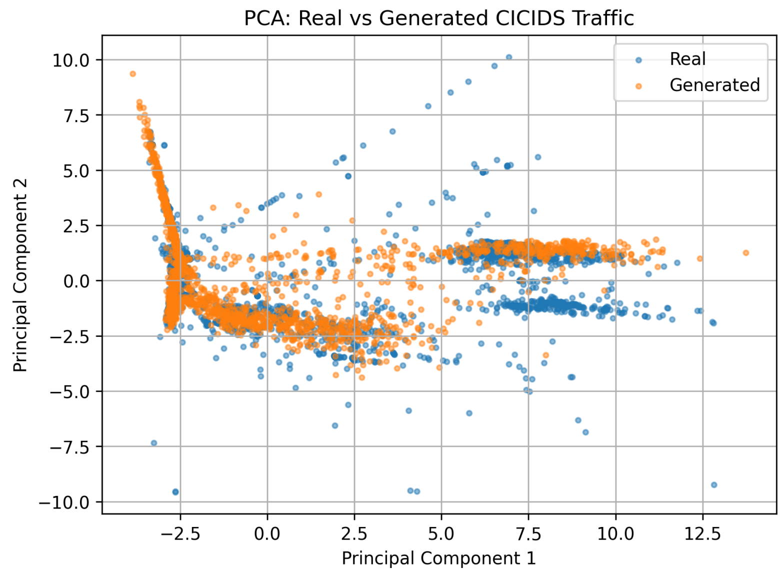

Figure 8: PCA Comparison of Real and Generated Traffic

The PCA projection demonstrated that generated samples occupied a similar region of feature space to real traffic samples. Quantitative evaluation produced a mean feature difference of 0.185 and a standard deviation difference of 0.444. These results suggest that the GAN successfully captured many characteristics of the original traffic distribution while maintaining variability among generated samples.

5. Part 2.3: Creative AI – Broccoli Sketch Generation

Dataset Exploration

The final experiment used the broccoli category from the Google Creative Lab QuickDraw dataset. Each sample is represented as a 28×28 grayscale sketch created by human users.

The objective was to train a DCGAN capable of generating realistic broccoli sketches from random noise.

Results and Discussion

The DCGAN successfully learned several common visual characteristics of broccoli drawings, including rounded crowns and vertical stems. Generated sketches displayed recognisable broccoli-like shapes, although some outputs remained blurry or incomplete.

Quantitative evaluation produced a simplified FID-style score of **302.72**, which was considerably higher than the score obtained for the medical imaging task. This result suggests that modelling hand-drawn sketches presents additional challenges due to variations in drawing style, stroke placement and abstraction.

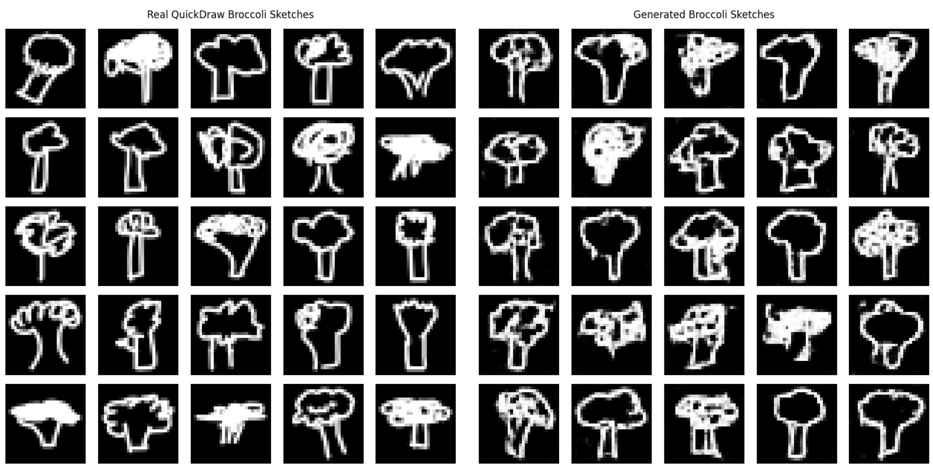

Figure 9: Real vs Generated Broccoli Sketches

Despite these limitations, the experiment demonstrated the potential of GANs for creative applications. Similar techniques are increasingly used in digital art, game design and content generation systems.

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6. Conclusion

This assignment investigated the use of Generative Adversarial Networks across synthetic, medical, cybersecurity and creative domains. The synthetic experiments demonstrated how GANs learn complex data distributions and how architectural modifications can improve performance. The PneumoniaMNIST experiment showed that DCGANs can generate realistic chest X-ray images with reasonable similarity to the original dataset, the cybersecurity experiment demonstrated that GANs can model network traffic distributions for analysis and augmentation purposes, and the QuickDraw experiment highlighted the potential of GANs in creative applications, although generating high-quality sketches remains challenging. Overall, the results show that GANs are versatile models capable of learning from both image and tabular data; while training stability and output quality remain important challenges, GANs continue to be a powerful approach for synthetic data generation across a wide range of application areas.

References

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